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Interdisciplinary Project Report

on

"SMART BLIND STICK"

Submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Interdisciplinary Project under the Innovation & Design Thinking Laboratory, Department of Basic Sciences.

by

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CERTIFICATE

This is to certify that the File Structures mini project entitled "SMART BLIND STICK " has been successfully carried out by Sanu Kumar(1CR25IS148), Sarah Fathima Shaik(1CR25IS149), Archana PM(1CR25AD015) and Arnav M(1CR25AD016), Bonafide students of **CMR Institute of Technology.**

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Interdisciplinary Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences.**

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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1.

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ACKNOWLEDGEMENT

I sincerely express my gratitude to Dr. Sanjay Jain, Principal, CMR Institute of Technology, Bangalore, for providing a supportive academic environment.

I extend my thanks to Dr.Fazlur Rahman, HoD, Dept. of Chemistry, CMRIT for his valuable guidance and support.

I am especially grateful to my internal guide, Dr. Prashant, Asst. Professor, Dept. of Chemistry for his constant encouragement and guidance throughout this project.

I also thank all the faculty members, non-teaching staff, and others who contributed directly or indirectly to the successful completion of this work.

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Abstract

The Smart Blind Stick is an assistive device designed to help visually impaired individuals navigate safely and independently. The project uses an ultrasonic sensor integrated with an Arduino Mega 2560 microcontroller to detect obstacles in front of the user. The ultrasonic sensor continuously measures the distance between the stick and nearby objects by transmitting and receiving ultrasonic waves. When an obstacle is detected within a predefined range, the system activates a buzzer to alert the user through sound signals. The frequency of the buzzer increases as the obstacle becomes closer, enabling better awareness of surrounding hazards. The device is portable, cost-effective, and energy efficient, making it suitable for daily use. The project aims to improve mobility, safety, and confidence for visually impaired individuals while reducing the risk of accidents. The implementation demonstrates the practical application of embedded systems, sensors, and real-time obstacle detection technology in assistive healthcare devices. The developed system provides reliable performance and can be further enhanced with additional features such as GPS tracking, voice guidance, and vibration feedback

Keywords:

Smart Blind Stick, Arduino Mega 2560, Ultrasonic Sensor, Obstacle Detection, Assistive Technology, Embedded System

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Chapter 1

INTRODUCTION

Visual impairment creates significant challenges in safe and independent navigation. Blind individuals often depend on traditional walking sticks, which cannot detect obstacles beyond physical contact range. This limitation increases the risk of accidents, collisions, and injuries in crowded or unfamiliar environments. The objective of this project is to develop a Smart Blind Stick using Arduino Mega 2560 and ultrasonic sensor technology to assist visually impaired individuals in obstacle detection and navigation. The system uses ultrasonic waves to identify nearby obstacles and alerts the user using a buzzer. The project focuses on providing a portable, low-cost, energy-efficient, and user-friendly assistive device that improves mobility and safety. The developed prototype demonstrates the practical implementation of embedded systems and sensor-based real-time obstacle detection technology.

1.1 Brief History of Blind Assistance Devices

Traditional blind sticks were primarily designed for physical surface detection and support. Over time, electronic assistive devices were introduced using sensors and microcontrollers to improve navigation and safety for visually impaired individuals. Modern smart sticks use ultrasonic sensors, GPS, IoT, and artificial intelligence technologies for advanced obstacle detection and navigation assistance.

1.2 Modern Smart Blind Stick System

Modern smart blind stick systems combine sensors, microcontrollers, and alert mechanisms to provide real-time obstacle detection. Ultrasonic sensors are commonly used because of their accuracy, affordability, and low power consumption. These systems improve user independence, safety, and mobility in both indoor and outdoor environments.

Chapter 2

Problem Statement

2.1 Description

Visually impaired individuals face difficulties in detecting obstacles that are beyond the reach of traditional walking sticks. Existing solutions may be expensive, less portable, or difficult to use. There is a need for a simple, affordable, and reliable assistive system that can detect nearby obstacles and provide immediate alerts to the user

2.2 Challenge Statement

To design and implement a low-cost Smart Blind Stick using Arduino Mega 2560 and ultrasonic sensor technology capable of detecting nearby obstacles and providing real-time warning alerts through a buzzer system for safe navigation.

Chapter 3

3.1 Design Thinking Process

The design thinking process was followed to develop the Arthritis-Friendly Pill Bottle in a systematic and user-centered manner. The steps involved are as follows:

Empathize:

Discussions with visually impaired individuals and observations of navigation challenges revealed difficulties in obstacle detection, mobility, and safety during movement.

Define:

The major requirements identified were:

- Accurate obstacle detection
- Fast response time
- Portable design
- Low power consumption
- Easy usability

Ideate:

Different obstacle detection methods such as infrared sensors, ultrasonic sensors, and camera-based systems were analyzed. The ultrasonic sensor-based solution was selected because of its simplicity, accuracy, affordability, and reliability.

Prototype:

A hardware prototype consisting of Arduino Mega 2560, HC-SR04 ultrasonic sensor, buzzer, and power supply was developed and assembled on a blind stick

Test:

The prototype was tested in indoor and outdoor environments. The system successfully detected nearby obstacles and provided warning alerts with satisfactory accuracy and response time

3.2 Methodology

- The ultrasonic sensor continuously transmits ultrasonic waves.
- Reflected waves from nearby obstacles are received by the sensor.
- Arduino Mega 2560 calculates the distance between the obstacle and the stick.
- If the obstacle is within a predefined range, the buzzer is activated.
- The buzzer frequency increases as the obstacle distance decreases.
- (Working procedure flowchart with description)

3.3 Prototype Description

3.3.1 Materials Used

a) Arduino Mega 2560

- Main microcontroller board
- Controls sensor and buzzer operations
- Operates at 5V

b) HC-SR04 Ultrasonic Sensor

- Detects nearby obstacles using ultrasonic waves
- Detection range: 2 cm to 400 cm

c) Active Buzzer

- Provides audio warning alerts
- Activated when obstacles are detected

d) Jumper Wires

- Used for electrical connections between components

e) Power Bank / Battery

- Supplies power to the Arduino and sensor system

f) Blind Stick Structure

- Supports mounting of electronic components

3.3.2 System Diagram

Circuit Connections

Component	**Arduino Mega 2560 Pin**
HC-SR04 VCC	5V
HC-SR04 GND	GND
HC-SR04 TRIG	Pin 9
HC-SR04 ECHO	Pin 10
Buzzer +	Pin 8
Buzzer -	GND

System Working Description

The ultrasonic sensor continuously scans for nearby obstacles. The sensor sends distance information to the Arduino Mega 2560. Based on the measured distance, the Arduino activates the buzzer to alert the user. The closer the obstacle, the faster the buzzer sound frequency.

Chapter 4

Implementation

Arduino Code

```
const int trigPin = 9;
const int echoPin = 10;
const int buzzer = 8;
long duration;
int distance;
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
  if(distance > 90 || distance <= 0){
     noTone(buzzer);
  }
  else if(distance > 60){
     tone(buzzer,2000);
     delay(100);
     noTone(buzzer);
```

```
    delay(700);
  }
  else if(distance > 35){
    tone(buzzer,2500);
    delay(80);
    tone(buzzer,3000);
    delay(80);
    noTone(buzzer);
    delay(300);
  }
  else if(distance > 15){
    tone(buzzer,3200);
    delay(50);
    tone(buzzer,3800);
    delay(50);
    noTone(buzzer);
    delay(100);
  }
  else{
    tone(buzzer,4000);
    delay(20);
    tone(buzzer,3500);
    delay(20);
  }
}
```

Chapter 5

Results and Analysis

User Testing & Feedback

Sample:

Participants: 10 students and 2 faculty members.

Quantitative Results:

Obstacle detection range achieved up to 200 cm.

Fast response time during testing.

Reliable obstacle detection accuracy in indoor environments.

Low power consumption observed during operation.

Qualitative Feedback:

Users found the system simple and effective.

The buzzer alert mechanism was clearly audible.

The prototype improved awareness of nearby obstacles.

Hardware Project

Photographs of the hardware setup and output are included. The prototype successfully detected obstacles and generated buzzer alerts during testing. The system demonstrated stable performance and effective obstacle warning capability.

The results demonstrate that the project objectives have been successfully achieved.

Chapter 6

Conclusion & Future Work

The Smart Blind Stick using Arduino Mega 2560 effectively improves obstacle detection and navigation assistance for visually impaired individuals. The project provides a low-cost, portable, and reliable assistive solution that enhances user safety and mobility. The ultrasonic sensor-based detection system successfully identifies nearby obstacles and alerts the user through audio signals.

Future Work:

- GPS tracking integration
- Voice guidance system
- Vibration alert mechanism
- IoT-based emergency communication
- AI-based object recognition
- Rechargeable battery management system

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